25/ $\text{Thermometer}_{t=0} = 5°C$

$\text{Room}_{t=0} = 22°C = T_A$

$\text{Thermometer}_{t=1} = 12°C$

$\text{Thermometer}_{t=x} = 21.9°C$

$$\frac{dT}{dt} = K(T - T_A)$$

$$dT\left(\frac{1}{T - T_A}\right) = K\,dt$$

$$\int dT\left(\frac{1}{T - T_A}\right) = \int K\,dt$$

$$\ln|T - T_A| = Kt$$

$$T(t) = T_A + Ce^{kt}$$

$$5° = 22° + Ce^{k(0)}$$

$$5° = 22° + C$$

$$-17° = C$$

$$T(t) = 22° - 17°e^{kt}$$

$$12° = 22° - 17°e^{k}$$

$$-10° = -17°e^{k}$$

$$\frac{-10}{-17} = e^{k}$$

$$\ln(10/17) = k$$

2S| $T(t) = 22° - 17°e^{-0.5306t}$

$21.9° = 22° - 17°e^{-0.5306t}$

$-0.1° = -17°e^{-0.5306t}$

$\ln\left(\frac{0.1}{17}\right) = -0.5306t$

$-5.1357 = -0.53306t$

$\boxed{t = 9.6345 \text{ minutes}}$

1S| $4y'' + 25y = 0 \quad y(0) = 3.0 \quad y'(0) = -2.5$

$\cos(2.5x), \sin(2.5x)$

$y = \cos(2.5x)$ $\quad y = \sin(2.5x)$

$y' = -2.5\sin(2.5x)$ $\quad y' = 2.5\cos(2.5x)$

$y'' = -6.25\cos(2.5x)$ $\quad y'' = -6.25\sin(2.5x)$

$4(-6.25\cos(2.5x)) + 25\cos(2.5x) = 0$

$-25\cos(2.5x) + 25\cos(2.5x) = 0$

$0 = 0$ ✓

$4(-6.25\sin(2.5x)) + 25\sin(2.5x) = 0$

$-25\sin(2.5x) + 25\sin(2.5x) = 0$

$0 = 0$ ✓

1S/ Proportionality:

$$\frac{\cos(2.5x)}{\sin(2.5x)} = \cot(2.5x) \neq \text{Constant}$$

Not Constant => Not proportional => linearly independent

IVP:

$y = C_1 y_1 + C_2 y_2 = C_1 \cos(2.5x) + C_2 \sin(2.5x)$

for $y(0) = 3$, $y'(0) = -2.5$

$$y(0) = C_1 \cos(2.5(0)) + C_2 \sin(2.5(0)) = 3$$

$$3 = C_1(1)$$

$$C_1 = 3$$

$$y' = -7.5\sin(2.5x) + 2.5C_2\cos(2.5x)$$

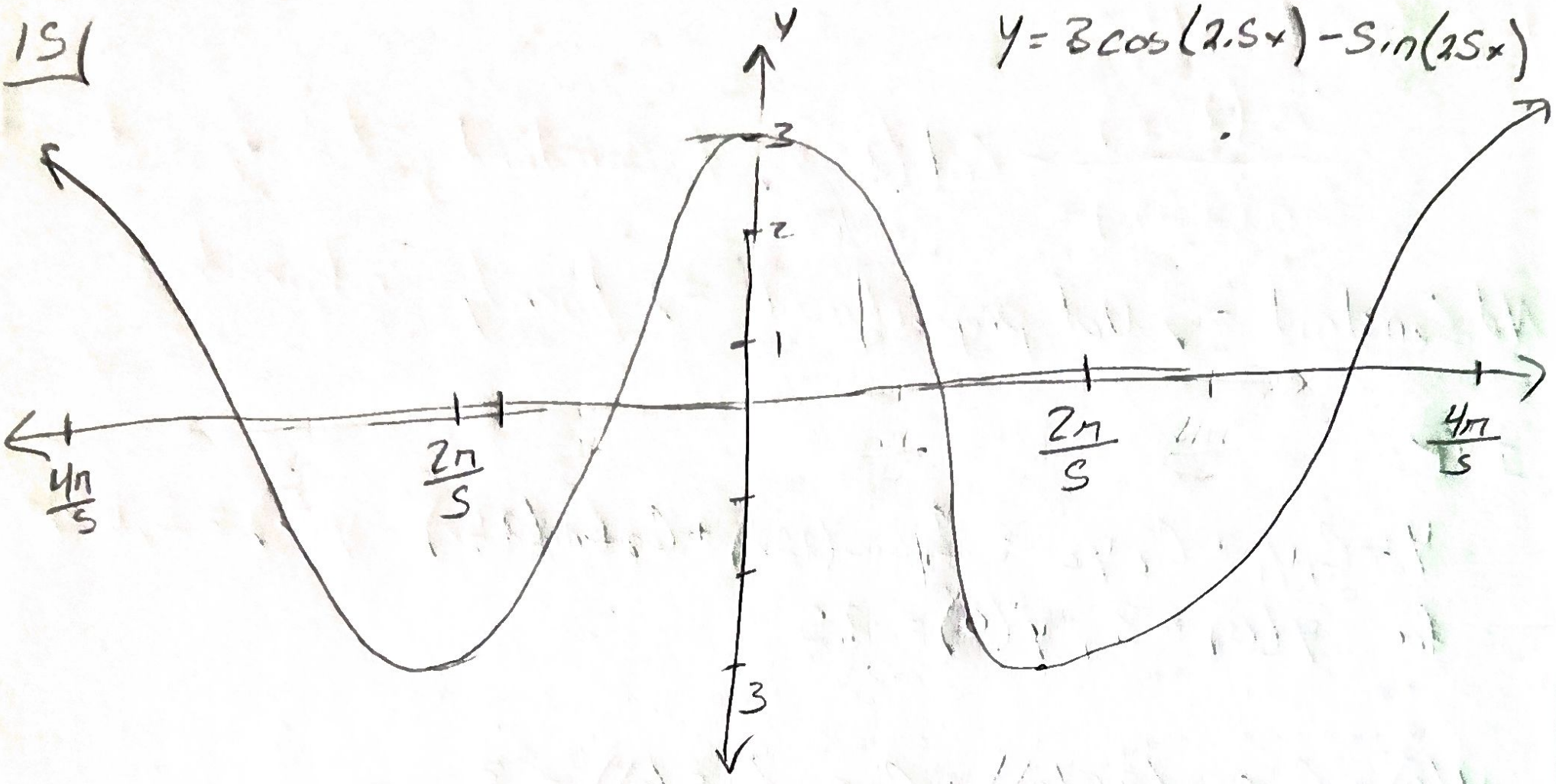
$$y'(0) = -7.5\sin(2.5(0)) + 2.5C_2\cos(2.5(0)) = -2.5$$

$$2.5C_2 = -2.5$$

$$C_2 = -1$$

$$y = 3\cos(2.5x) - \sin(2.5x)$$

1S)



$y = 3\cos(2.5x) - 5.\sin(25x)$

5)

$$y'' + 2\pi y' + \pi^2 y = 0$$

$$\lambda^2 + 2\pi\lambda + \pi^2 = 0$$

$$(\lambda + \pi)(-\lambda + \pi) = 0$$

$$\lambda = -\pi$$

Real, double roots → Cas II

$$\boxed{y = (C_1 + C_2 x)e^{-\pi x/2}}$$

27| $y'' + 2y' + r^2 = 0$ $\quad y(0) = 4.5, \quad y'(0) = -4.5r - 1$

$$y = (C_1 + C_2 x)e^{-rx/2}$$

$$y' = C_2 e^{-rx} - r(C_1 + C_2 x)e^{-rx}$$

$$y(0) = 4.5 = (C_1 + C_2(0))e^{-r(0)/2}$$

$$= (C_1 + 0)\cdot e^0$$

$$4.5 = C_1$$

$$y'(0) = -4.5r - 1 = C_2 e^{-r(0)} - r(4.5 + C_2(0))e^{-r(0)}$$

$$-4.5r - 1 = C_2 - r(4.5)$$

$$-1 = C_2$$

$$\boxed{y = (4.5 - 1x)e^{-rx/2}}$$

3(

$$y'' + 3y' + 2y = 12x^2$$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$(\lambda + 2)(\lambda + 1) = 0$$

$$\lambda = -2, -1 \implies y_h = C_1 e^{-x} + C_2 e^{-2x}$$

$y_p = k_2x^2 + k_1x + k_0$, because $12x^2$ is order 2

$y_p' = 2k_2x + k_1$

$y_p'' = 2k_2$

$$2k_2 + 3(2k_2x + k_1) + 2(k_2x^2 + k_1x + k_0)$$

$$= 2k_2 + 6k_2x + 3k_1 + 2k_2x^2 + 2k_1x + 2k_0$$

$$= 2k_2x^2 + (6k_2 + 2k_1)x + 2k_2 + 3k_1 + 2k_0 = 12x^2$$

$2k_2 = 12 \rightarrow k_2 = 6$

$6(6) + 2k_1 = 0 \rightarrow k_1 = 36/2 = -18$

$2(6) + 3(-18) + 2k_0 = 0 \rightarrow 12 + (-54) + 2k_0 = 0$

$k_0 = 42/2 = 21$

$y_p = 6x^2 - 18x + 21$

$$\boxed{y = C_1 e^{-x} + C_2 e^{-2x} + 6x^2 - 18x + 21}$$